

Handling Single Variable

1. Overview

Understanding the reasons for missing data is key to handling missing data and minimizing bias in analysis results. Generally, there are four categories of missing data mechanisms:

- **MCAR – Missing Completely at Random**

There is no missingness pattern in the data. The probability of missing data is the same for all observation units.

- **MAR – Missing at Random**

Missing patterns can be explained from observed data. The probability of missingness depends solely on the available information.

- **Missingness that depends on unobserved predictors**

Missingness is no longer random because it depends on information not recorded in the data.

- **Missingness that depends on the missing value itself**

Missingness depend on the (potentially missing) variable itself

Approaches to handling missing data can generally be divided into two main strategies, namely data deletion (discard data/deletion) and data imputation (retain data/imputation).

In this case study, the approach chosen to handle missing values is an imputation strategy using Regression Imputation, which is implemented through two methods, namely Deterministic imputation and Random imputation.

2. Background

The *airquality* dataset records daily air quality measurements in New York (May-September). We want to build a regression model to explain *Ozone* level using other variables such as *Solar.R*, *Wind*, and *Temp*. But we face the problem of missing data in the outcome (*Ozone*) and in one of the predictors (*Solar.R*).

We will demonstrate how to handle single variable missing data (missing in one variable at a time) in two different ways:

1. Discard / Delete Data (Complete Case Analysis)
2. Imputation
 - Using deterministic regression imputation
 - Using random regression imputation

Then we compare the evaluation of the regression models and see which method has the best performance

3. Exploration Data

An initial examination revealed the following pattern of missing data in the dataset, which consists of a total of 153 observations

Column	Missing Data	Percentage Missing
Ozone	37	24.18%
Solar.R	7	4.58%
Wind	0	0.00%
Temp	0	0.00%
Month	0	0.00%
Day	0	0.00%

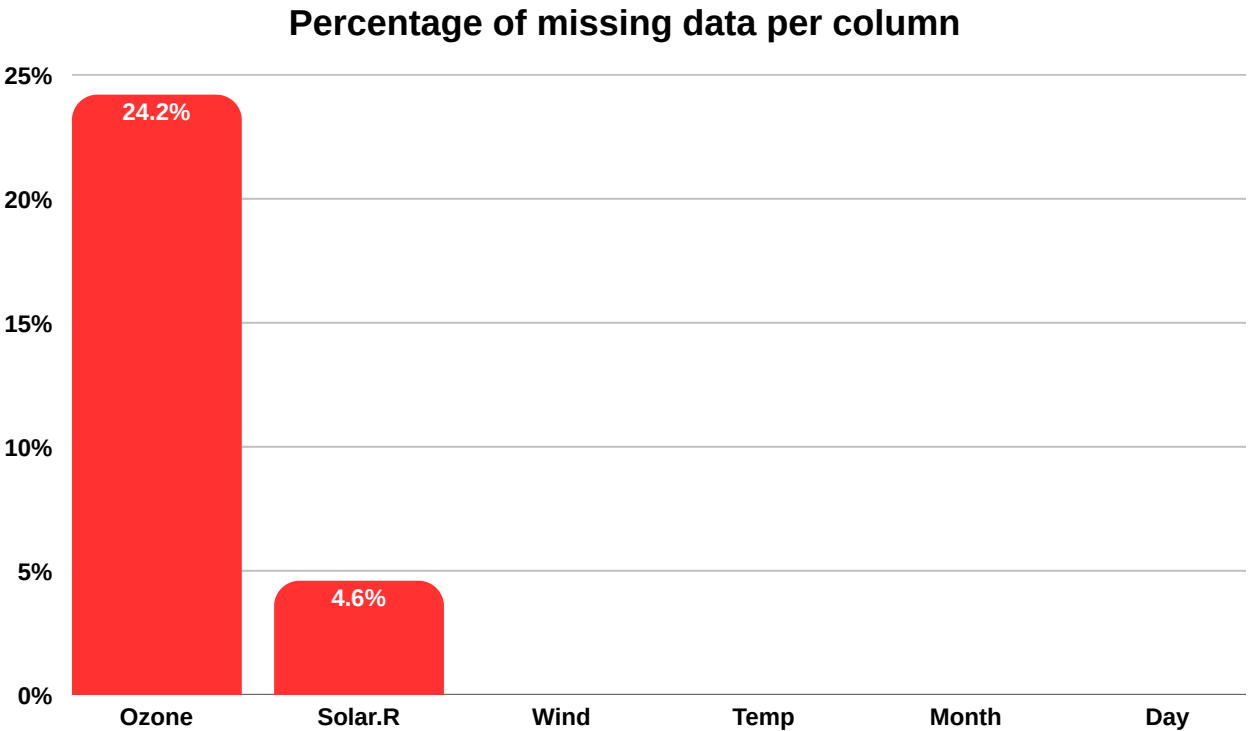


Figure 3.1 Percentage of missing data per column

Ozone has approximately 24% missing data, while Solar.R has approximately 5%. These variables make it a good candidate predictor for imputing Ozone. Because the missingness pattern of Wind, Temp, Month, and Day is fully recorded for Ozone, it can be explained by these fully observed variables. The missingness is assumed to be **MAR (Missing at Random)**, not MCAR—making the strategy of **retaining data through imputation** more appropriate than simply dropping rows.

For analysis purposes, the variable *Solar.R* was excluded from the list of predictors from the outset because it itself had missing values—to ensure consistency in comparisons between the *Ozone* treatment methods (models built with *Wind*, *Temp*, *Month*, and *Day*).

4. Missing Data Handling Method

4.1. Complete Case Analysis(CCA)

The simplest method is to remove all rows containing missing values (listwise deletion). This approach is suitable when missingness is assumed to be MCAR, but risks wasting valuable information if the missingness is actually MAR.

	Rows	Column
Original Data	153	7
After CCA	111	7

Dropping rows leaves 111 of the 153 observations — a loss of about 27% of the data, mostly due to missing values in Ozone.

4.2 Deterministic Regression Imputation

A regression model was constructed using complete rows (non-missing ozone) with predictors Wind, Temp, Month, and Day. This model was then used to predict ozone values in the missing rows, and these predictions were used directly as imputation values.

4.3 Random Regression Imputation

Similar to deterministic imputation, but each predicted value is added to a random error taken from a normal distribution with a standard deviation of the model residual ($\sigma \approx 21.4$), so that the imputation results retain the natural variability of the data.

5. Results and Comparison of Model Performance

5.1 R-squared Comparison

Method	R-squared
Complete Case Analysis	0.6085
Deterministic Regression Imputation	0.6494
Random Regression Imputation	0.6112

5.2 Comparison of Standard Error Coefficients

Coefficient	CCA	Det. Regression	Random Regression
Intercept	23.877	17.504	18.929
Wind	0.655	0.481	0.520
Temp	0.263	0.196	0.212
Month	1.501	1.172	1.268
Day	0.233	0.171	0.185

The R-squared of the Complete Case Analysis model tends to be the lowest of the three methods, while its standard error is the highest. Deterministic regression imputation produces the highest R-squared and the smallest standard error because it removes the natural variability of missing values (the predicted values are exactly on the regression line). Random regression imputation falls somewhere in between—it adds back some of the missing variability, thus better representing the uncertainty of the original data than the deterministic version.

6. Ozone Value Distribution: Observations and Imputation Results

The following is a visual comparison between the directly observed Ozone distribution and the distribution of imputed values from the two regression methods:

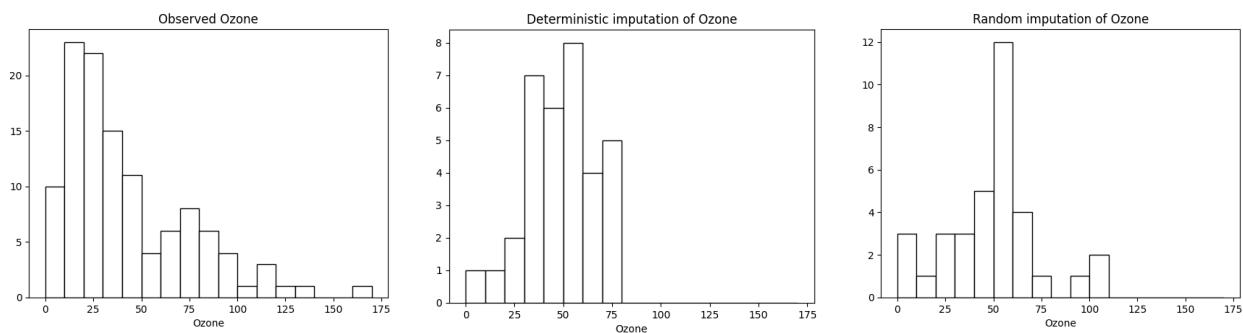


Figure 6.1 – Observed Ozone distribution, deterministic imputation results, and random imputation results

Random regression imputation produces a more dispersed (varied) distribution of values than deterministic imputation, which tends to produce more centralized values because it relies solely on point predictions from the regression line without a random error component.

7. Conclusion & Recommendation

- Data issuance for **Ozone** (24.18%) and **Solar.R** (4.58%) was handled using three approaches: Complete Case Analysis, Deterministic Regression Imputation, and Random Regression Imputation.
- **Complete Case Analysis** discarded 27% of the data rows, resulting in a model with the lowest R-squared (0.6085) and the highest standard error—risking bias if the missingness is actually MAR.
- **Deterministic Regression Imputation** provided the highest R-squared (0.6494) and the smallest standard error, but underestimated the true variability because the imputed values precisely followed the prediction line.
- **Random Regression Imputation** (R-squared 0.6112) provided a more realistic compromise: it retained data uncertainty through a random error component, so the estimated standard error was not underestimated as with the deterministic method.
- **Recommendation:** For the most numerically precise coefficient estimates, deterministic imputation is superior; However, for honest statistical inference regarding uncertainty (including the calculation of valid standard errors and confidence intervals), random regression imputation is recommended because it does not artificially reduce data variability.